Lab 3.1 FMEA document generation

IES | 30 minutes | LO1, LO2, LO3

All records are fictional training data. They describe no actual facility, person or organisation. Values are teaching assumptions, not approved design criteria. Do not use outputs to operate, maintain or approve real equipment.

Inputs and preparation

Open the mock-data folder: system-description.txt, fmea-template.csv, rating-scales.txt. Use prompts.md for copy and paste. Save your reviewed work under outputs. Earlier lab outputs are optional; this folder provides standalone inputs.

Procedure

1. Read the system description and classroom rating scale. Open fmea-template.csv in Excel and save a working copy as an Excel workbook in approved storage. Keep the source CSV unchanged.

2. Format the data as a table with headers. Use Copilot to review function, failure mode, effect, cause and current control. Keep current controls separate from proposed actions.

3. Inspect S, O and D ratings and add a justification for each. In cell J2 enter =G2*H2*I2 and fill down to J4, or use the equivalent structured-reference formula for the table.

Lab 3.1 continued

4. Sort RPN largest to smallest while selecting the whole table. Verify F01=192, F02=189 and F03=175 manually. Also review severity 9 in F02 even though it is not the largest RPN.

5. Ask for action, owner, due date and evidence required for each row. Do not reduce residual ratings simply because an action is proposed.

6. In Word, draft a one-page FMEA review summary with scope, top findings, unresolved evidence, action ownership and a blank professional review decision. Save the workbook and summary.

Check before moving on

RPNs are 192, 189 and 175; complete rows remain aligned after sorting.

Ratings have stated justifications and high severity is reviewed.

Summary remains a draft awaiting review.

Access or source problems

Use approved paired access or the local source files. Record unavailable live capabilities as not executed. If an attachment is not supported, paste the short synthetic text or reference its approved SharePoint copy. Do not use personal accounts to bypass controls.

Lab 3.1 prompts and evidence

Main prompt

Using SYS-01 and the supplied training scale, review the FMEA rows. Preserve distinctions among failure mode, effect, cause and current control. Explain rating assumptions, compute RPN, and propose actions with owners and verification evidence. Treat ratings as provisional workshop judgements. Review high severity separately. Never imply a proposed action has already reduced risk.

Verification prompt

Audit the draft against the named source files. List each claim, source location, uncertainty and correction. Recalculate key values independently. Return a revised draft and change log; do not approve the engineering deliverable.

Expected output example

Illustrative teaching example, not a recorded AI run. F02 reverse flow: S9 x O3 x D7 = 189. Monthly inspection is the current control. Review backflow protection is a proposal. Residual rating remains unassessed until effectiveness evidence is reviewed.

Retain your evidence

Keep the source versions, prompt, tool, output, corrections, independent checks and reviewer decision. Record any capability or permission test that was not executed. Use templates/prompt-log.csv. Do not publish real workplace data.